

Julian Pacheco

📍 Portland, OR | @ apacheco26@outlook.com | 🌐 julian-pacheco.com | 📄 github.com/apacheco26

SUMMARY

Data analyst with an MS in Data Science, focused on turning clinical and research datasets into dashboards, reports, and descriptive and predictive analyses that non-technical teams and clinical stakeholders act on. Four plus years at OHSU using FlowJo, GraphPad Prism, and Excel daily for biomedical data analysis, and independently building self-serve analytics, reporting, classification models, and production-grade data pipelines beyond formal job requirements.

TECHNICAL SKILLS

Lab & Research Analysis: FlowJo, GraphPad Prism, Excel

Analytics and Reporting (BI dashboarding): Plotly / Dash, R / Shiny, Grafana, descriptive statistics, window functions, statistical and longitudinal modeling

Machine Learning: scikit-learn (Pipeline, ColumnTransformer), NumPy, Matplotlib, Seaborn, TF-IDF / CountVectorizer, logistic regression, GridSearchCV, classification metrics (accuracy, F1, precision/recall)

Languages: Python, SQL, R

Data Pipelines (cloud SQL warehousing; supporting the analytics layer): Airflow, dbt, DuckDB, PostgreSQL, S3, pandas, ETL design, star schema modeling

Tooling and Deployment: Git, Docker, Railway, Flask, Quarto, Beekeeper Studio, Jupyter Notebook, VS Code

PROFESSIONAL EXPERIENCE

Vaccine & Gene Therapy Institute (OHSU)

Beaverton, OR

Data Analyst / Research Assistant

2021 – Present

- Partnered with multidisciplinary academic, clinical, and international teams to turn complex biomedical datasets into findings supporting translational research and a peer reviewed publication.
- Collaborated with research partners across 9+ institutions in 3 countries, coordinating multi-site data and sample management for translational HIV research.
- Integrated longitudinal datasets spanning flow cytometry, viral load, cytokine, and molecular assays into unified structures for treatment response and viral reservoir analysis.
- Analyzed high dimensional, multiparameter flow cytometry and longitudinal immune monitoring data in FlowJo and GraphPad Prism, and independently built reproducible analytical workflows and reporting pipelines, beyond formal job requirements, from human and nonhuman primate studies.

Bureau of Land Management

Lakeview, OR

Biological Field Technician (Contract via Oregon State University)

March 2021 – August 2021

- Curated and integrated geospatial, telemetry, vegetation, and environmental datasets for a multi year restoration study; analysis supported land management decisions and a peer reviewed publication.

ANALYTICS & REPORTING PROJECTS

County Affordability & Automation Risk (Capstone) / Python, SQL, PostgreSQL (Railway)

- Built the ETL pipeline and Railway-hosted PostgreSQL warehouse ingesting five public datasets (Census ACS, BLS, BEA, O*NET) across three agencies for a 3,143-county, 15-year (2008–2023) panel of 47,145 county-year observations, as the data engineer on a two-person capstone.
- Designed the SQL schema and ERD, and diagnosed a multicollinearity problem in the task-mix features (VIF ~27), fixing it by converting task-group counts to within-county proportions (VIF near 1).
- Ran the statistical analysis linking county task-group composition to housing affordability, informing a model comparison (Random Forest test R2 0.840 vs. 0.610 for OLS/Ridge) that found job automatability an independent, non-linear predictor of affordability beyond poverty and unemployment.
- Built the project's data-flow diagrams and poster figures presented for the DATA 510 Data Science Capstone.

News Credibility Classifier / Python, scikit-learn

- Built binary and multi-class text classifiers predicting news credibility, combining TF-IDF vectorization with logistic regression in a scikit-learn Pipeline / ColumnTransformer across a 44,906-article labeled corpus and a separate 6-class truthfulness dataset.

- Tuned vectorizer n-gram range, document-frequency thresholds, and regularization strength via GridSearchCV, reaching 99.6% test accuracy on binary classification and 0.58 macro F1 on the hardest 2-class collapse of the multi-class task.
- Ran a label-shuffling sanity check to rule out data leakage, confirming the pipeline dropped to 50.1% accuracy (chance level) once the true label signal was removed.

MortalityTrends Analytics Dashboard / *R / Shiny, PostgreSQL*

- Shipped an interactive R / Shiny dashboard (choropleth maps, disparity views) for exploring 25 years of county level CDC WONDER mortality data across 3,000+ counties.
- Modeled a county disparity index via PERCENT_RANK and categorized causes of death into 18 analytical buckets for descriptive reporting.
- Caught and fixed a data quality defect where a population denominator break across dataset versions inflated dementia rates after 2021.
- Built the underlying scraping and staging pipeline pulling county level CDC WONDER data (1999 to 2024) with session management, rate limit handling, and fetch logging.

Beyond the Grid: Energy Investment Dashboard / *Python, PostgreSQL, Plotly / Dash*

- Deployed an interactive Plotly / Dash application ranking state renewable investment against climate potential, using a composite alignment metric.
- Engineered the alignment metric with PERCENT_RANK window functions partitioned by year, with correlations computed in SQL via CORR().
- Documented coverage gaps and backfill decisions (Alaska exclusion, PVGIS limits) as explicit, reproducible data limitations for report consumers.
- Built the ETL pipeline ingesting seven public sources (EIA, NOAA, NSRDB, BEA, Census, OWID, PVGIS) into a PostgreSQL warehouse.

Prague Metro Ridership Dashboard / *Airflow, dbt, DuckDB, PostgreSQL*

- Built Grafana dashboards answering ridership questions with pivot style SQL over a dbt star schema (fact table plus zone, fare type, date, and hour of day dimensions).
- Revised the fact grain after design review feedback to better support ad hoc reporting questions.
- Diagnosed and fixed reporting-affecting production bugs: a dbt aggregation grain bug freezing tap counts to a constant, a DuckDB S3 write error, and Airflow connection errors.
- Built and extended the underlying Dockerized Airflow pipeline (snapshot DAGs to an S3 lake, staging loads, a MongoDB feed) that keeps the dashboards current.

EDUCATION

Willamette University	Salem, OR
<i>M.S., Data Science</i>	<i>August 2026</i>
Portland State University	Portland, OR
<i>B.S., Biology</i>	<i>June 2017</i>

PUBLICATIONS

Takata H, Mitchell JL, **Pacheco J**, et al. (2023). An active HIV reservoir during ART is associated with maintenance of HIV-specific CD8+ T cell magnitude and short-lived differentiation status. *Cell Host & Microbe* 31(9), 1494–1506.e4.